



Fundamentals of Optics

Lecturer

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Q&A time: 2-4 pm on Monday

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Rules

- Final exam: 40%
- Mid-term exam: 30%
- Homework: 20%
- Attendance: 10%

Reference books

Optics, by Eugene Hecht
光学, 赵凯华、钟锡华

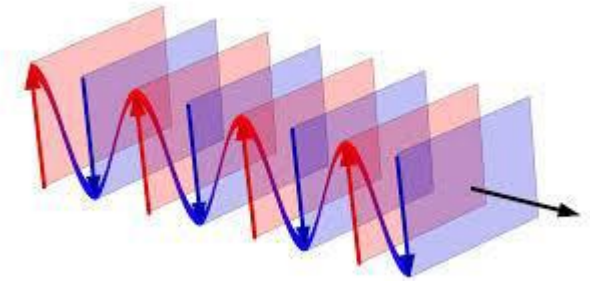
CHAPTER 2

ELECTROMAGNETIC THEORY, PHOTONS, AND LIGHT

Review

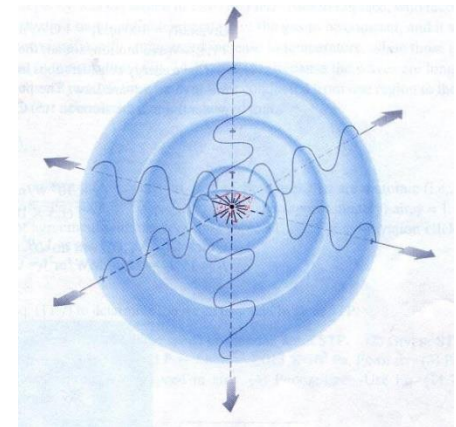
➤ Plane wave

$$\psi(\vec{r}) = Ae^{i(\vec{k} \cdot \vec{r} \mp \omega t)}$$



➤ Spherical wave

$$\psi(r, t) = \left(\frac{\mathcal{A}}{r}\right) \cos k(r \mp \omega t)$$



➤ Basic parameters of a harmonic wave

Phase velocity $v = \frac{\omega}{k}$

Propagation vector $k = 2\pi/\lambda$

Period $\tau = \lambda/v$

➤ One-dimensional **differential wave equation**

$$\frac{\partial^2 \psi}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2}$$

➤ Three-dimensional **differential wave equation**

$$\nabla^2 \psi = \frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2}$$

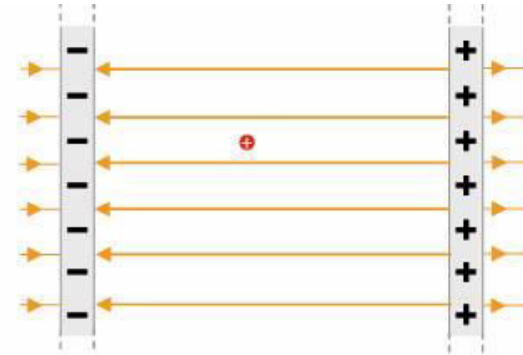
Wave-particle manifestations

- ❑ Classical electrodynamics unalterably leads to the picture of a continuous transfer of energy by ways of electromagnetic waves.
- ❑ Quantum electrodynamics describes electromagnetic interactions and the transport of energy in terms of massless elementary “particles” known as photons.
- ❑ Geometrical optics: the wavelength of light is small in comparison to the size of the apparatus (lenses, mirrors, etc.)

2.1 Basic laws of electromagnetic theory

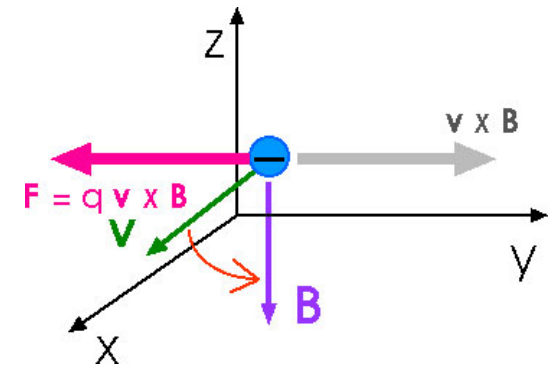
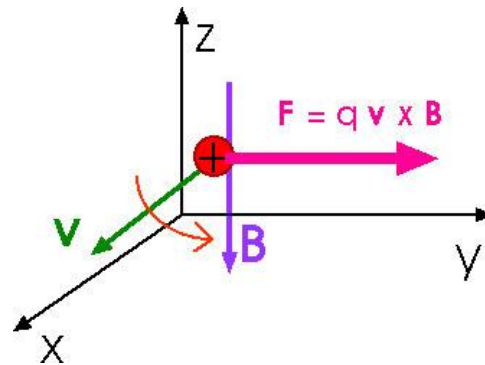
□ Electric field $\vec{E}$

$$\vec{F}_E = q \cdot \vec{E}$$



□ Magnetic induction or magnetic field $\vec{B}$

$$\vec{F}_M = q \cdot \vec{v} \times \vec{B}$$



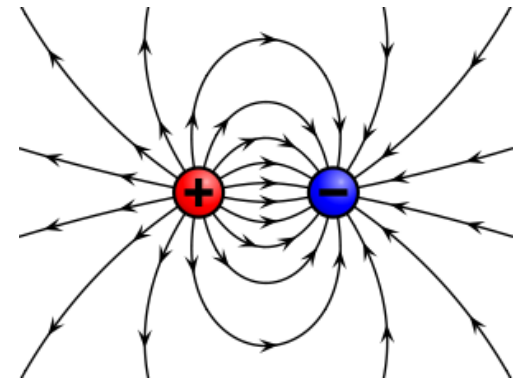
Left-hand rule

Source

❑ Electric fields are generated by

Electric charges

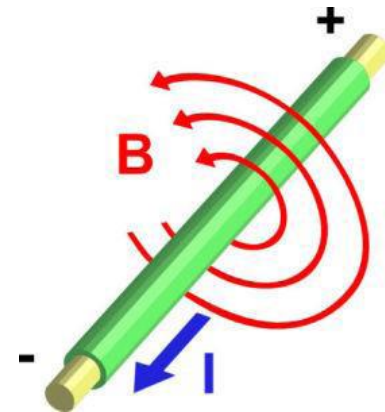
Time-varying magnetic fields



❑ Magnetic fields are generated by

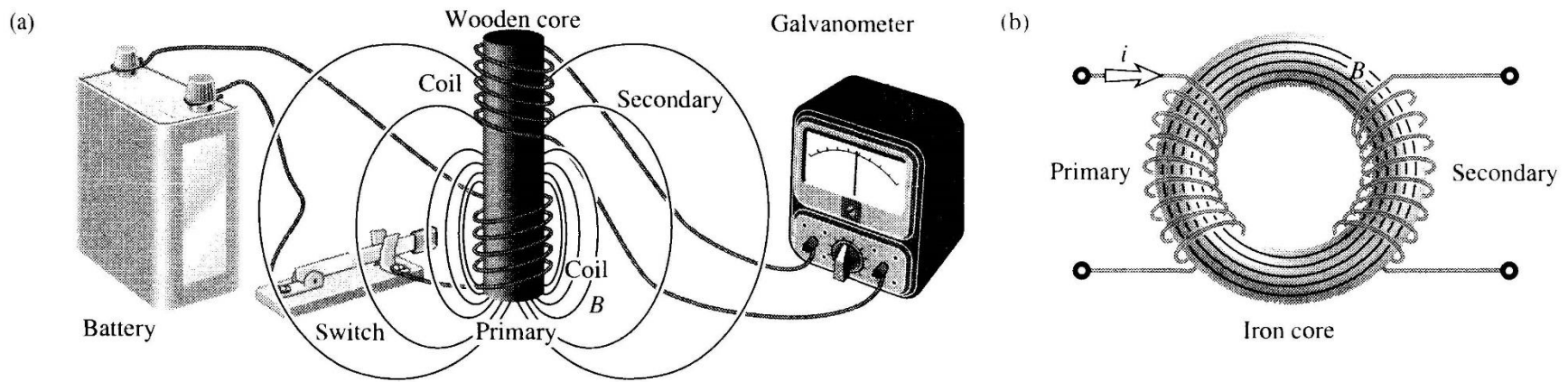
Electric currents

Time-varying electric fields



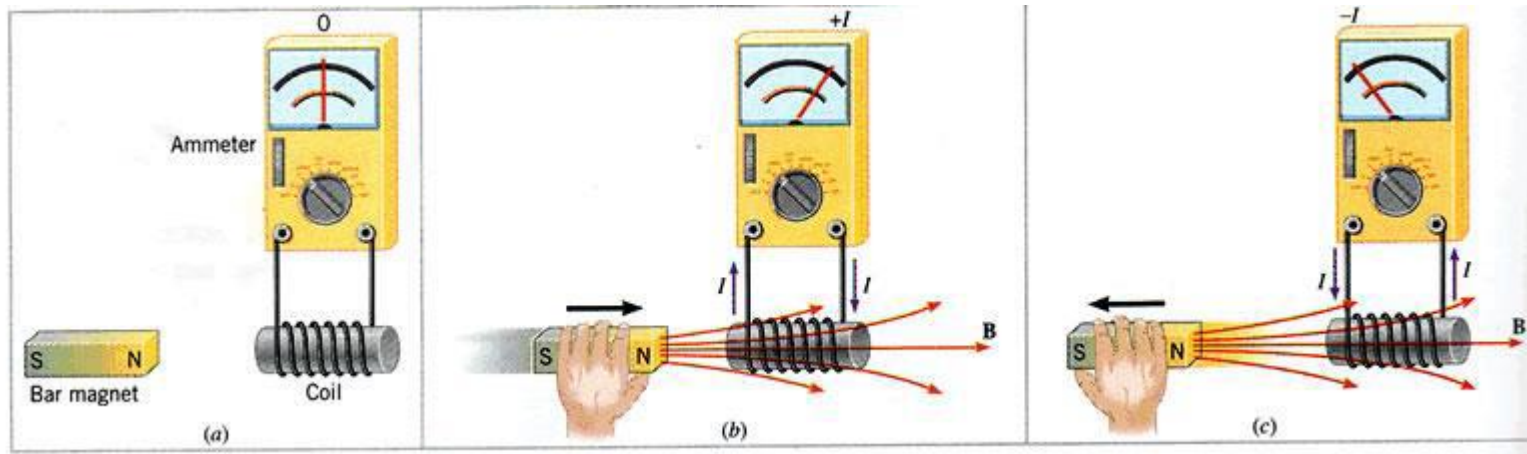
2.1.1 Faraday's Induction Law

□ A changing magnetic field generated a current



(a) The start of a current in one coil produces a time varying magnetic field that induces a current in the other coil. (b) An iron core couple the primary coil to the secondary

□ By thrusting a magnet into a coil, Faraday showed that there is a voltage known as the induced electromotive force (emf, 感应电动势) across the coil's terminals.



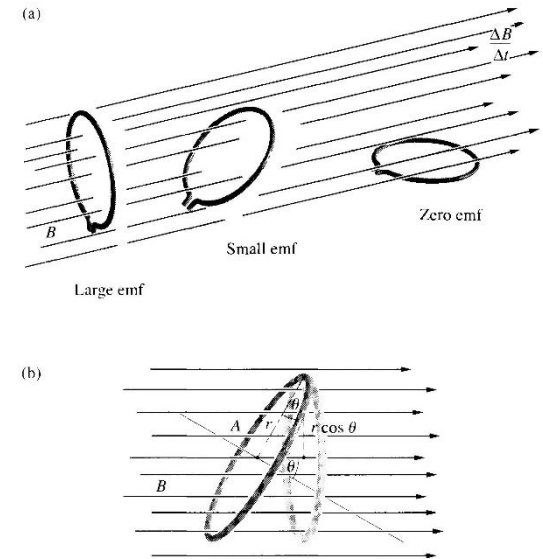
The induced emf depends on the rate-of-change of B through the coil and not on B itself.

- When the B is changing, the induced emf is proportional to the area A of the loop penetrated perpendicularly by the field.

$$emf \propto A_{\perp} \Delta B / \Delta t$$

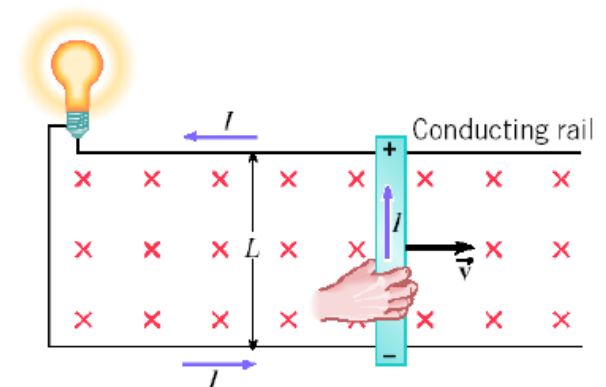
$$emf = - \frac{d\phi}{dt}$$

$$d\phi = \int dA_{\perp} dB$$



- When the field is constant, the induced emf is proportional to the rate-of-change of the perpendicular area penetrated

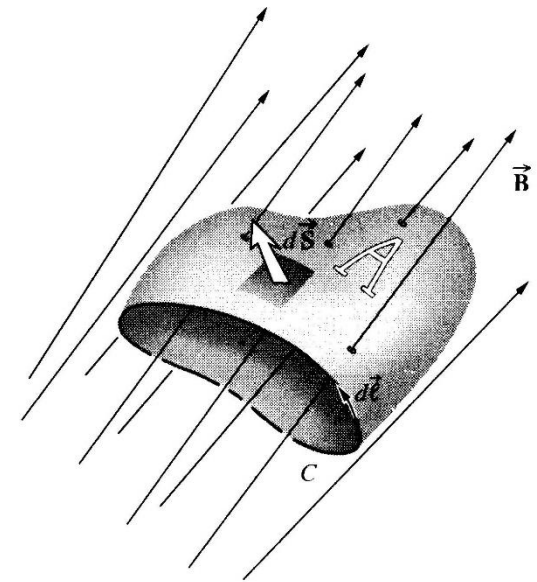
$$emf \propto B \Delta A_{\perp} / \Delta t$$



❑ The emf depends on the rate-of-change of both $A_{\perp}$ and B

❑ Flux of the magnetic field (Φ_M)

$$\Phi_M = B_{\perp} A = BA_{\perp} = BA \cos \theta$$



❑ More generally,

$$\Phi_M = \iint_A \vec{B} \cdot d\vec{S}$$

❑ The induced emf:

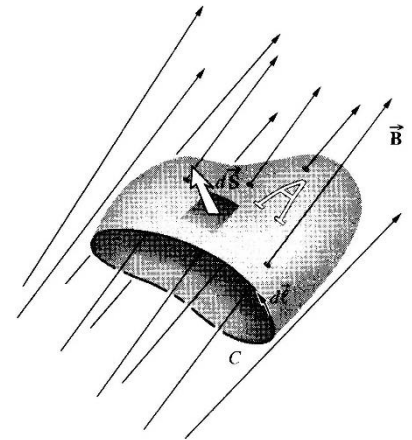
$$\text{emf} = -\frac{d\Phi_M}{dt}$$

- emf: a potential difference (potential-energy difference per unit charge; work done per unit charge; force per unit charge times distance, electric field times distance):

$$emf = \oint_C \vec{E} \cdot d\vec{\ell}$$

- Faraday's Induction Law

$$\oint_C \vec{E} \cdot d\vec{\ell} = -\frac{d}{dt} \iint_A \vec{B} \cdot d\vec{S}$$



- As to electromagnetic waves traveling in space where there are no wire loop, so the induction law:

$$\oint_C \vec{E} \cdot d\vec{\ell} = -\iint_A \frac{\partial \vec{B}}{\partial t} \cdot d\vec{S}$$

A time-varying magnetic field will have an electric field associated with it.

2.1.2 Gauss's Law- Electric

- ❑ The volume flux (Av) through both end surfaces is equal in magnitude.

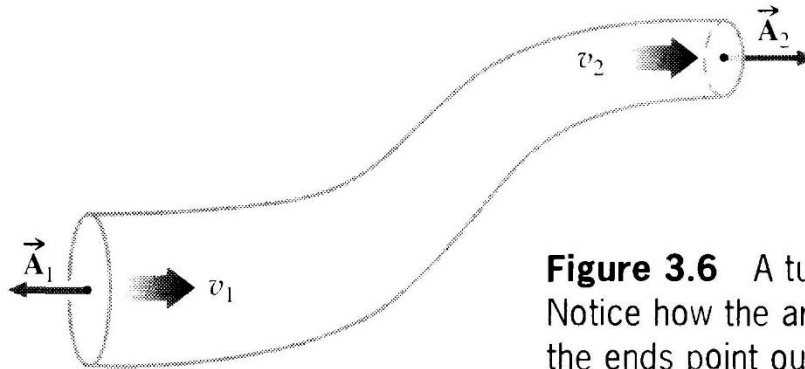
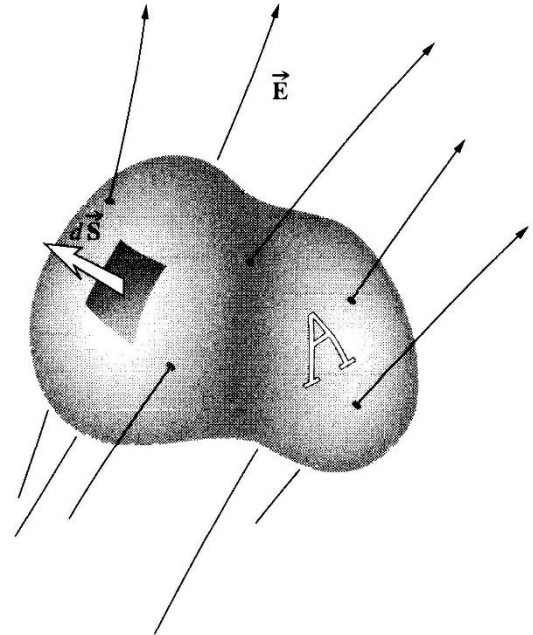


Figure 3.6 A tube of fluid flow. Notice how the area vectors on the ends point outward.

- ❑ When a small pipe is inserted into the region either sucking out fluid (a sink) or delivering fluid (a source), the net flux will be nonzero.

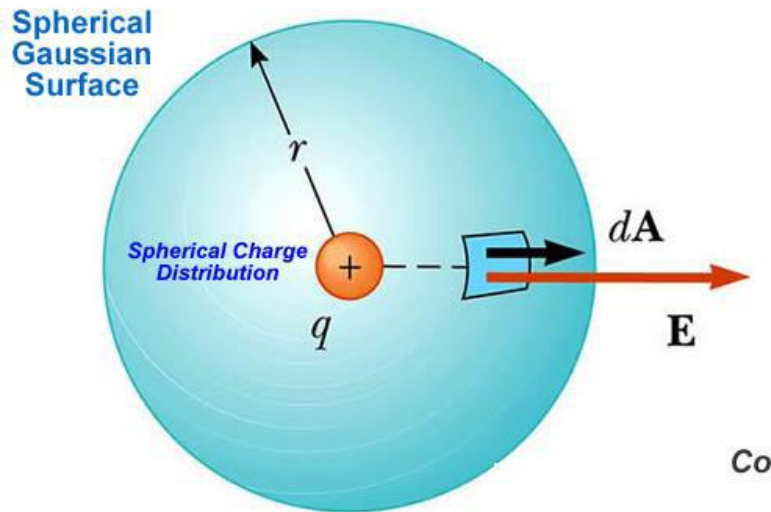
- The flux of the electric field through A:

$$\underline{\Phi_E = \oiint_A \vec{E} \cdot d\vec{S}}$$



- When there are no sources or sinks of the electric field within the region encompassed by the closed surface, the net flux through the surface equals zero.

- ❑ What happened in the presence of the internal sources and sinks:
- ❑ Consider spherical surface of radius r centered on a positive point-charge (q) in vacuum.



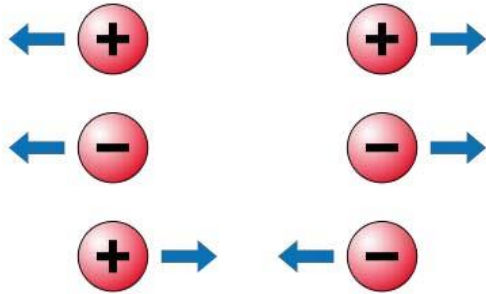
Gauss's Law

$$\oint_A \vec{E} \cdot d\vec{S} = \frac{1}{\epsilon_0} \iiint_V \rho dV$$

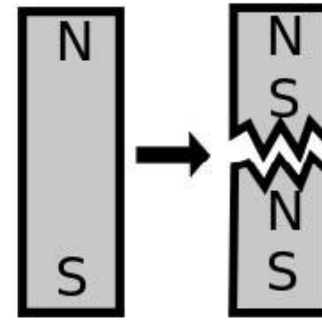
ϵ_0 : electric permittivity (电容率) of free space and equals to $8.8542 \times 10^{-12} \text{C}^2/(\text{N} \cdot \text{m}^2)$

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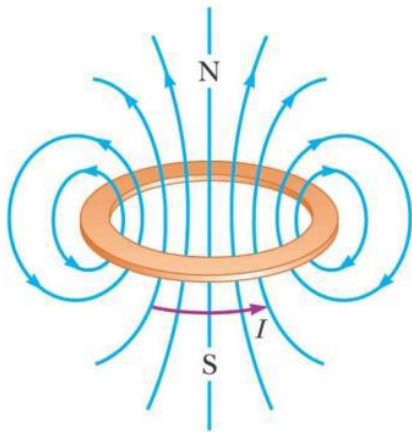
2.1.3 Gauss's Law- Magnetic



Electric charge



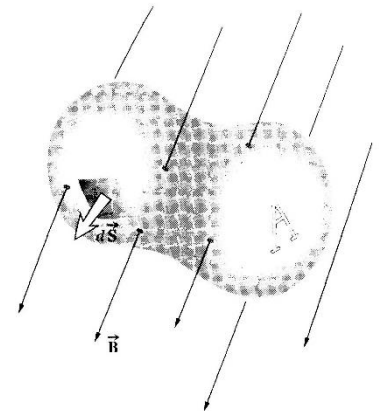
Magnetic poles



Magnetic fields can be described in terms of current distributions

Magnetic equivalent of Gauss's Law

$$\Phi_M = \oint_A \vec{B} \cdot d\vec{S} = 0$$



2.1.4 Ampere's Circuital Law

安培环路定律

- Imagine a straight current-carrying wire in the vacuum and the circular B-field surrounding it.
- From experiments

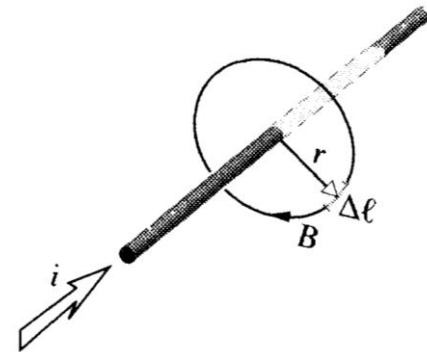
$$B = \mu_0 i / 2\pi r$$

Ampere's Law

$$\oint_C \vec{B} \cdot d\vec{\ell} = \mu_0 \sum i$$

Ampere's Law in terms of the current density

$$\oint_C \vec{B} \cdot d\vec{\ell} = \mu_0 \iint_A \vec{J} \cdot d\vec{S}$$



The $\vec{B}$ -field surrounding a current-carrying wire.

Ampere's Law: is it the truth?

□ Consider a capacitor charging:

As to area A_1 : the above
Ampere's law is truth.

As to area A_2 : it is not truth

When charging or discharging a capacitor, one can measure a B -field around the wire and in the region between its plates

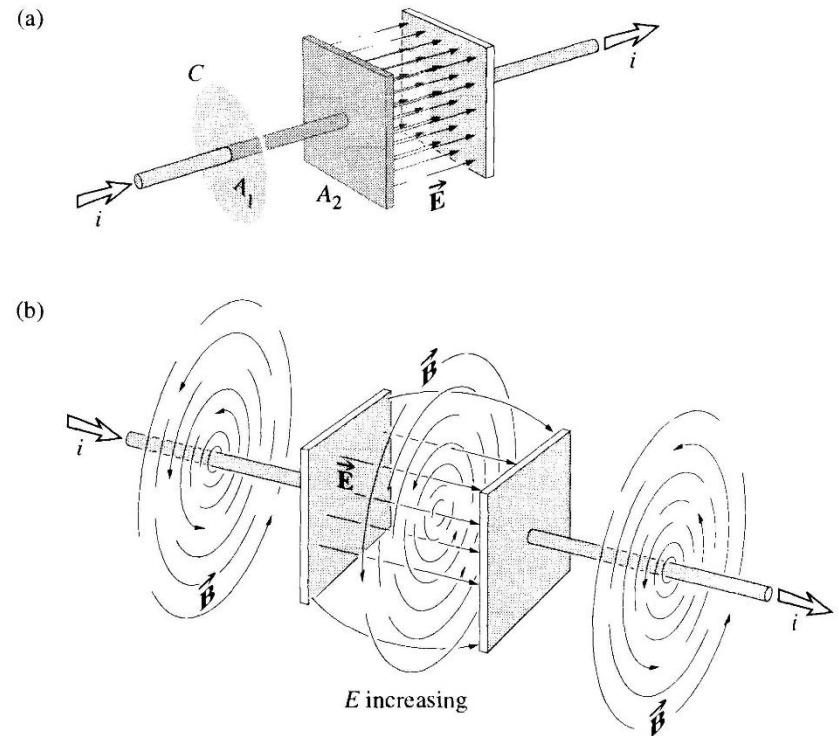
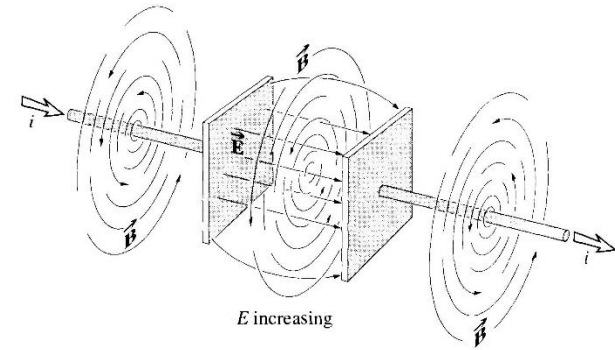


Figure 3.11 (a) Ampère's Law is indifferent to which area A_1 or A_2 is bounded by the path C . Yet a current passes through A_1 and not through A_2 , and that means something is very wrong. (b) $\vec{B}$ -field concomitant with a time-varying $\vec{E}$ -field in the gap of a capacitor.

□ The electric field of a capacitor

$$\underline{E = \frac{Q}{\epsilon A}}$$

A: the area of each plate;
Q: the charge on it.



□ As the charge varies, the electric field changes

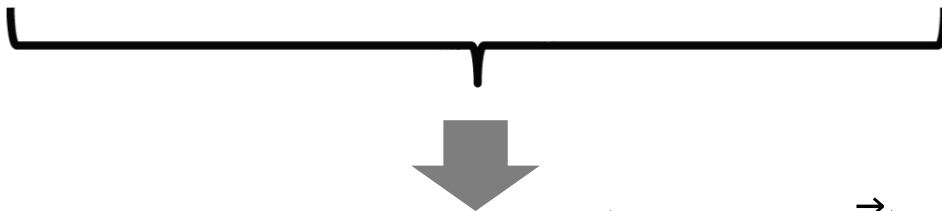
$$\epsilon \frac{\partial E}{\partial t} = \frac{i}{A}$$

□ Maxwell's contribution: define a displacement current density

位移电流密度

$$\vec{J}_D \equiv \epsilon \frac{\partial \vec{E}}{\partial t}$$

- ❑ Combine Ampere's law and Maxwell's displacement current density:

$$\oint_C \vec{B} \cdot d\vec{\ell} = \mu_0 \iint_A \vec{J} \cdot d\vec{S} \quad \vec{J}_D \equiv \varepsilon \frac{\partial \vec{E}}{\partial t}$$


$$\oint_C \vec{B} \cdot d\vec{\ell} = \mu_0 \iint_A \left(\vec{J} + \varepsilon \frac{\partial \vec{E}}{\partial t} \right) \cdot d\vec{S}$$

μ_0 : the permeability (磁导率) of free space and it is defined as $4\pi \times 10^{-7} \text{ N} \cdot \text{s}^2 / \text{C}^2$

- ❑ When $\vec{J} = 0$, a time-varying $\vec{E}$ -field will be accompanied by a $\vec{B}$ -field.

2.1.5 Maxwell's equations

□ Faraday's Induction Law

$$\oint_C \vec{E} \cdot d\vec{\ell} = -\frac{d}{dt} \iint_A \vec{B} \cdot d\vec{S}$$

□ Gauss's Law- Electric

$$\oiint_A \vec{E} \cdot d\vec{S} = \frac{1}{\epsilon_0} \iiint_V \rho dV$$

□ Gauss's Law- Magnetic

$$\oiint_A \vec{B} \cdot d\vec{S} = 0$$

□ Ampere-Maxwell's Law

$$\oint_C \vec{B} \cdot d\vec{\ell} = \mu_0 \iint_A \left(\vec{J} + \epsilon \frac{\partial \vec{E}}{\partial t} \right) \cdot d\vec{S}$$

□ Maxwell's equations in free space

$$\epsilon = \epsilon_0; \mu = \mu_0; \text{ and } \rho = 0; J = 0$$

$$\oint_C \vec{E} \cdot d\vec{\ell} = -\iint_A \frac{\partial \vec{B}}{\partial t} \cdot d\vec{S}$$

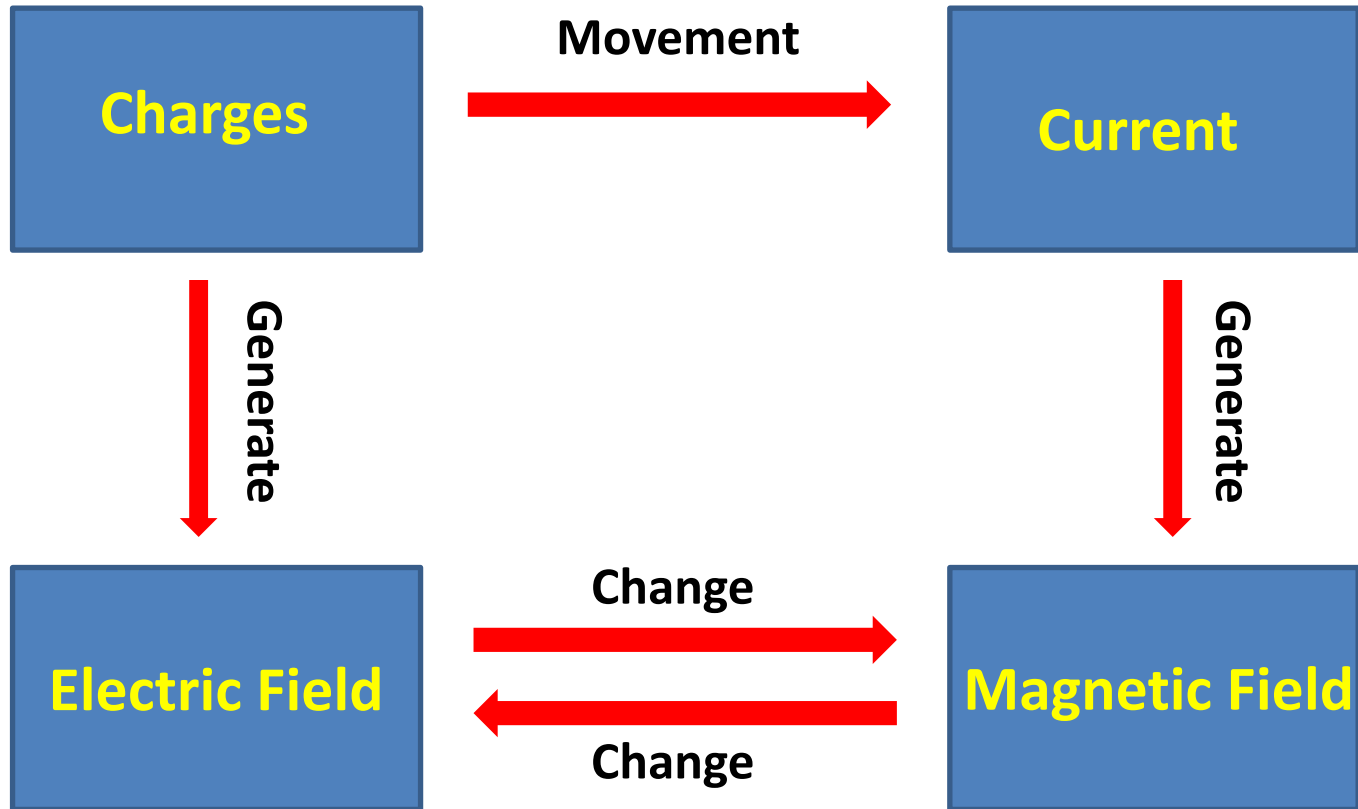
$$\oiint_A \vec{E} \cdot d\vec{S} = 0$$

$$\oiint_A \vec{B} \cdot d\vec{S} = 0$$

$$\oint_C \vec{B} \cdot d\vec{\ell} = \mu_0 \epsilon_0 \iint_A \frac{\partial \vec{E}}{\partial t} \cdot d\vec{S}$$

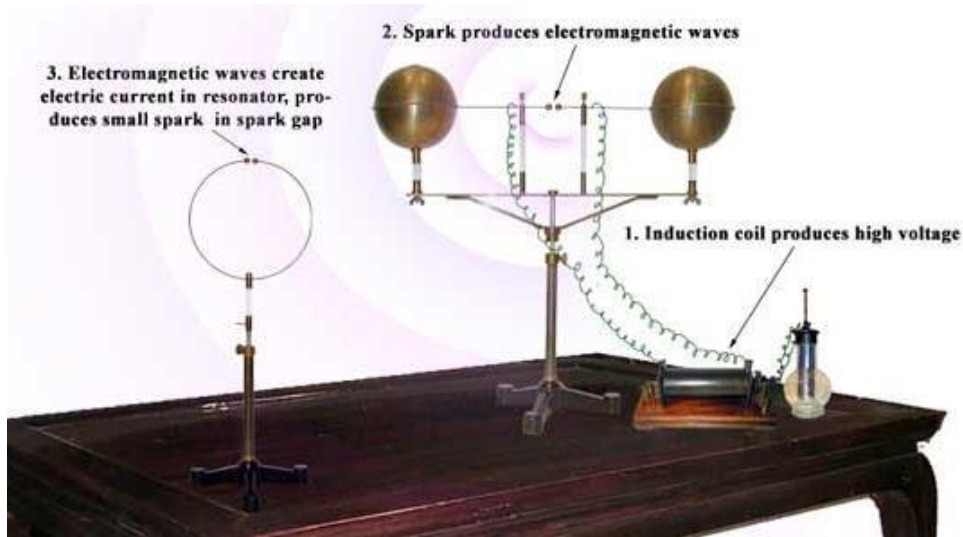
□ A time-varying magnetic field generates an electric field and a time-varying electric field generates a magnetic field.

Link between electric field and magnetic field



Hertz's Experiment

Video link



- ☐ Hertz said “It's of no use whatsoever, this is just an experiment that proves Maestro Maxwell was right.”
- ☐ Marconi and Popov did their greatest contribution ([wireless telegraph](#)) to the application of electromagnetic wave.

Review

□ Maxwell's equations in free space

$$\varepsilon = \varepsilon_0; \mu = \mu_0; \text{ and } \rho = 0; J = 0$$

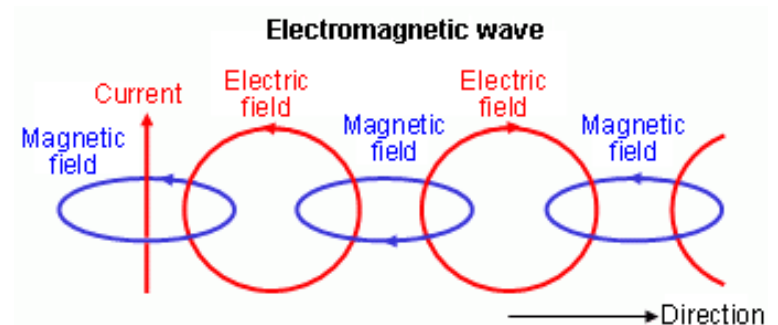
$$\oint_C \vec{E} \cdot d\vec{\ell} = -\frac{d}{dt} \iint_A \vec{B} \cdot d\vec{S}$$

$$\oiint_A \vec{E} \cdot d\vec{S} = 0$$

$$\oiint_A \vec{B} \cdot d\vec{S} = 0$$

$$\oint_C \vec{B} \cdot d\vec{\ell} = \mu_0 \varepsilon_0 \iint_A \frac{\partial \vec{E}}{\partial t} \cdot d\vec{S}$$

- *A time-varying magnetic field generates an electric field and a time-varying electric field generates a magnetic field.*



Maxwell's equations written in differential form

Based on Gauss's divergence theorem 高斯散度定理

Stokes's theorem 斯托克斯定理

$$\oint_C \vec{E} \cdot d\vec{\ell} = -\frac{d}{dt} \iint_A \vec{B} \cdot d\vec{S}$$

$$\oiint_A \vec{E} \cdot d\vec{S} = 0$$

$$\oiint_A \vec{B} \cdot d\vec{S} = 0$$

$$\oint_C \vec{B} \cdot d\vec{\ell} = \mu_0 \epsilon_0 \iint_A \frac{\partial \vec{E}}{\partial t} \cdot d\vec{S}$$

$$\oiint_A \vec{F} \cdot d\vec{S} = \iiint_V \nabla \cdot \vec{F} dV$$

$$\oint_C \vec{F} \cdot d\vec{\ell} = \iint_A \nabla \times \vec{F} \cdot d\vec{S}$$



$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\nabla \cdot \vec{E} = 0$$

$$\nabla \cdot \vec{B} = 0$$

$$\nabla \times \vec{B} = \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

$$\nabla \times \vec{E} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ E_x & E_y & E_z \end{vmatrix}$$

$$\nabla \cdot \vec{E} = \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z}$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$



$$\begin{aligned} \frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z} &= -\frac{\partial B_x}{\partial t} \\ \frac{\partial E_x}{\partial z} - \frac{\partial E_z}{\partial x} &= -\frac{\partial B_y}{\partial t} \\ \frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} &= -\frac{\partial B_z}{\partial t} \end{aligned}$$

$$\nabla \cdot \vec{E} = 0$$



$$\frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z} = 0$$

$$\nabla \cdot \vec{B} = 0$$



$$\frac{\partial B_x}{\partial x} + \frac{\partial B_y}{\partial y} + \frac{\partial B_z}{\partial z} = 0$$

$$\nabla \times \vec{B} = \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$



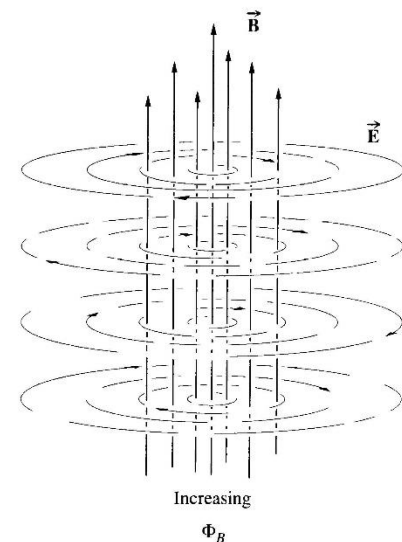
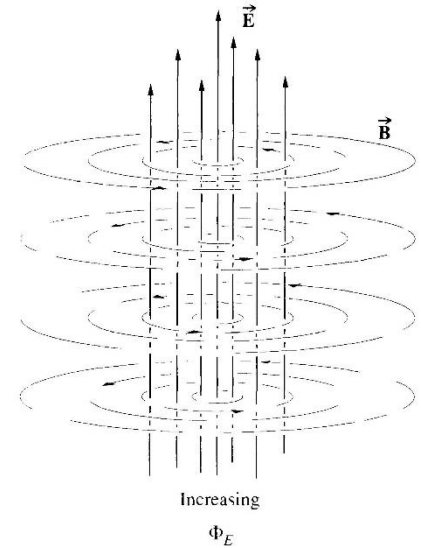
$$\begin{aligned} \frac{\partial B_z}{\partial y} - \frac{\partial B_y}{\partial z} &= \mu_0 \epsilon_0 \frac{\partial E_x}{\partial t} \\ \frac{\partial B_x}{\partial z} - \frac{\partial B_z}{\partial x} &= \mu_0 \epsilon_0 \frac{\partial E_y}{\partial t} \\ \frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y} &= \mu_0 \epsilon_0 \frac{\partial E_z}{\partial t} \end{aligned}$$

2.2 Electromagnetic waves

Maxwell's equation tell us:

□ A time-varying $\vec{E}$ -field generates a $\vec{B}$ -field, which is everywhere perpendicular to the direction in which $\vec{E}$ changes.

□ A time-varying $\vec{B}$ -field generates a $\vec{E}$ -field, which is everywhere perpendicular to the direction in which $\vec{B}$ changes.



Maxwell's equations in the form of two vector expressions

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon} \quad \nabla \cdot \vec{B} = 0 \quad \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad \nabla \times \vec{B} = \mu\sigma\vec{E} + \mu\epsilon\frac{\partial \vec{E}}{\partial t}$$

$$\begin{aligned} \nabla \times (\nabla \times \vec{B}) &= \mu\sigma(\nabla \times \vec{E}) + \mu\epsilon\frac{\partial}{\partial t}(\nabla \times \vec{E}) = -\mu\sigma\frac{\partial \vec{B}}{\partial t} - \mu\epsilon\frac{\partial^2 \vec{B}}{\partial t^2} \\ \nabla \times (\nabla \times \vec{B}) &= \nabla(\nabla \cdot \vec{B}) - \nabla^2 \vec{B} = -\nabla^2 \vec{B} \end{aligned} \quad \left. \vphantom{\begin{aligned} \nabla \times (\nabla \times \vec{B}) &= \mu\sigma(\nabla \times \vec{E}) + \mu\epsilon\frac{\partial}{\partial t}(\nabla \times \vec{E}) = -\mu\sigma\frac{\partial \vec{B}}{\partial t} - \mu\epsilon\frac{\partial^2 \vec{B}}{\partial t^2} \\ \nabla \times (\nabla \times \vec{B}) &= \nabla(\nabla \cdot \vec{B}) - \nabla^2 \vec{B} = -\nabla^2 \vec{B} \end{aligned}} \right\} \nabla^2 \vec{B} - \mu\sigma\frac{\partial \vec{B}}{\partial t} - \mu\epsilon\frac{\partial^2 \vec{B}}{\partial t^2} = 0$$

$$\begin{aligned} \nabla \times (\nabla \times \vec{E}) &= -\frac{\partial}{\partial t}(\nabla \times \vec{B}) = -\mu\sigma\frac{\partial \vec{B}}{\partial t} - \mu\epsilon\frac{\partial^2 \vec{E}}{\partial t^2} \\ \nabla \times (\nabla \times \vec{E}) &= \nabla(\nabla \cdot \vec{E}) - \nabla^2 \vec{E} \end{aligned} \quad \left. \vphantom{\begin{aligned} \nabla \times (\nabla \times \vec{E}) &= -\frac{\partial}{\partial t}(\nabla \times \vec{B}) = -\mu\sigma\frac{\partial \vec{B}}{\partial t} - \mu\epsilon\frac{\partial^2 \vec{E}}{\partial t^2} \\ \nabla \times (\nabla \times \vec{E}) &= \nabla(\nabla \cdot \vec{E}) - \nabla^2 \vec{E} \end{aligned}} \right\} \nabla^2 \vec{E} - \mu\sigma\frac{\partial \vec{B}}{\partial t} - \mu\epsilon\frac{\partial^2 \vec{E}}{\partial t^2} = \nabla\left(\frac{\rho}{\epsilon}\right)$$

In free space: $\epsilon = \epsilon_0$; $\mu = \mu_0$; and $\rho = 0$; $\sigma = 0$

$$\nabla^2 \vec{E} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2} \quad \nabla^2 \vec{B} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{B}}{\partial t^2}$$

Maxwell's equations for free space in the form of two vector expressions

$$\nabla^2 \vec{E} = \underbrace{\mu_0 \epsilon_0}_{\text{red wavy}} \frac{\partial^2 \vec{E}}{\partial t^2} \quad \nabla^2 \vec{B} = \underbrace{\mu_0 \epsilon_0}_{\text{red wavy}} \frac{\partial^2 \vec{B}}{\partial t^2}$$

The six components of the electromagnetic field $E_x, E_y, E_z, B_x, B_y, B_z$ obey the scalar differential wave equation

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} = \frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2}$$

$$v = \underbrace{1/\sqrt{\mu_0 \epsilon_0}}_{\text{red wavy}}$$

$$v = \frac{1}{\sqrt{\mu_0 \epsilon_0}} \approx 3 \times 10^8 \text{ m/s} = c$$

This theoretical value was in remarkable agreement with the previously measured speed of light determined by Fizeau.

2.2.1 Transverse waves

- Consider the simple case of a plane wave propagating in vacuum in the positive x-direction.

$$\vec{E} = \vec{E}(x, t)$$

- According to Maxwell's equation:

$$\nabla \cdot \vec{E} = 0 \rightarrow \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z} = 0$$

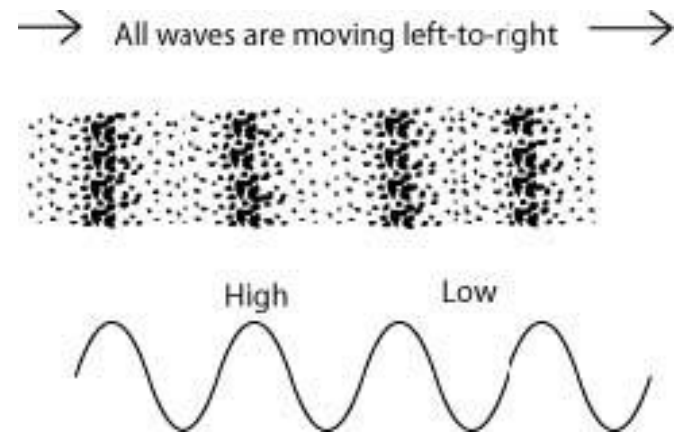
- We have

$$\frac{\partial E_x}{\partial x} = 0$$

- $E_x = \text{constant}$: it does not vary with x, so possibility cannot correspond to a traveling wave advancing in the positive x-direction.

- So $E_x = 0$

- The electromagnetic wave has no electric field component in the direction of propagation.



- ❑ To specify the wave, we first specify the moment-by-moment direction of $\vec{E}$.
- ❑ For simplify, we orient our coordinate axes to y-axis

$$\vec{E} = \hat{j}E_y(x, t)$$

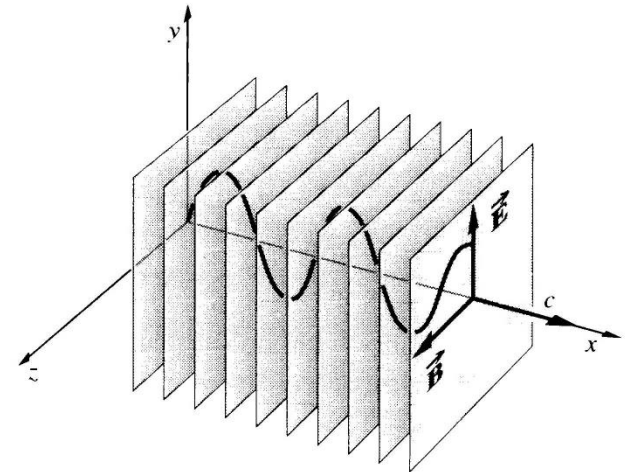
$$\frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z} = -\frac{\partial B_x}{\partial t}$$

$$\frac{\partial E_x}{\partial z} - \frac{\partial E_z}{\partial x} = -\frac{\partial B_y}{\partial t}$$

$$\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} = -\frac{\partial B_z}{\partial t}$$

$$\frac{\partial E_y}{\partial x} = -\frac{\partial B_z}{\partial t}$$

The time-dependent $\vec{B}$ -field can only have a component in the z-direction



- ❑ In free space, the plane electromagnetic wave is transverse.

Harmonic waves

$$E_y(x, t) = E_{0y} \cos \left[\omega \left(t - \frac{x}{c} \right) + \varepsilon \right]$$

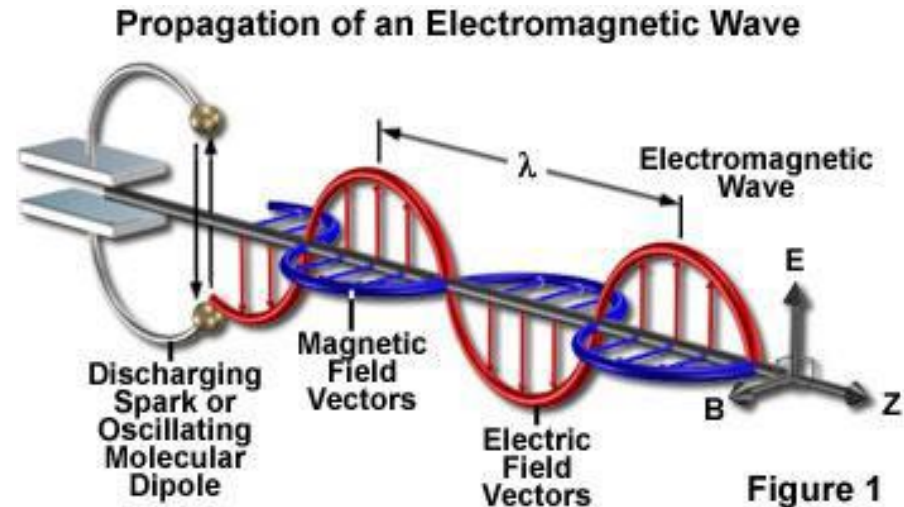
$$\frac{\partial E_y}{\partial x} = - \frac{\partial B_z}{\partial t}$$

$$\textcircled{B_z} = \int \frac{\partial E_y}{\partial x} dt$$

$$= - \frac{E_{0y}}{c} \int \sin \left[\omega \left(t - \frac{x}{c} \right) + \varepsilon \right] dt$$

$$= \frac{1}{c} E_{0y} \cos \left[\omega \left(t - \frac{x}{c} \right) + \varepsilon \right]$$

$$= \textcircled{\frac{E_y}{c}}$$



$$B_z = \frac{E_y}{c}$$

- $\vec{E}$ and $\vec{B}$ have the same time-dependence;
- $\vec{E}$ and $\vec{B}$ are in-phase at all points in space;
- $\vec{E}$ and $\vec{B}$ are mutually perpendicular;
- $\vec{E} \times \vec{B}$ points in the propagation direction.

2.3 Energy: The Poynting Vector

- Any electromagnetic wave exists within some region of space, and the field has energy it is a thing-in-itself.

Energy density (u): the radiant energy per unit volume

The energy density of the $\vec{E}$ -field:

$$u_E = \frac{\epsilon_0}{2} E^2 \quad u_E = \frac{\epsilon_0}{2} E^2$$

The energy density of the $\vec{B}$ -field:

$$u_B = \frac{1}{2\mu_0} B^2$$

Using $c = 1/\sqrt{\mu_0\epsilon_0}$, we have $u_E = u_B$ ✓ $u_B = \frac{1}{2\mu_0} B^2$

The total energy density of electromagnetic fields $(B = \frac{E}{c})$

$$\underline{u_{total}} = u_E + u_B = \epsilon_0 E^2 = \frac{1}{\mu_0} B^2$$

The transport of energy per unit time across a unit area.

$$\underline{S} = \frac{uc\Delta tA}{\Delta tA} = uc = \frac{1}{\mu_0} EB$$

Vector $\vec{S}$: energy flowing in the direction of the propagation of the wave

沿传播方向流动的能量.

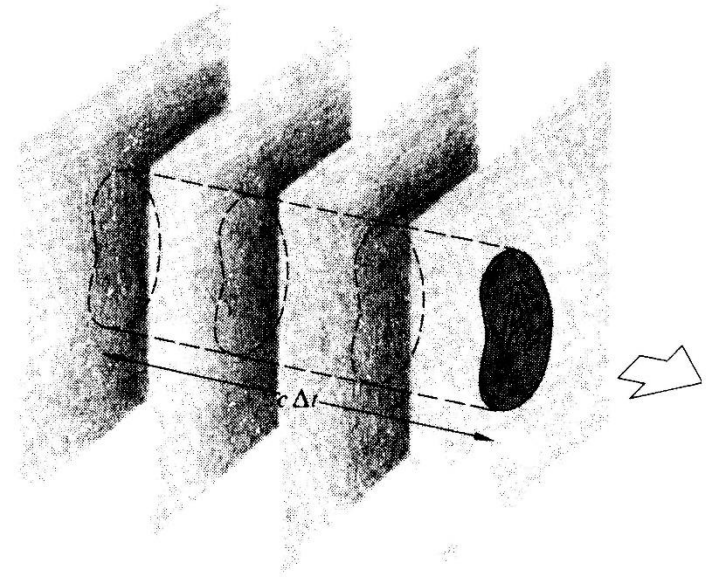
$$\vec{S} = \frac{1}{\mu_0} \vec{E} \times \vec{B}$$

$$\vec{S} = c^2 \epsilon_0 \vec{E} \times \vec{B}$$

坡印廷矢量.

The Poynting vector: the magnitude of $\vec{S}$ is the power per unit area crossing a surface whose normal is parallel to $\vec{S}$

与 $\vec{S}$ 垂直的平面



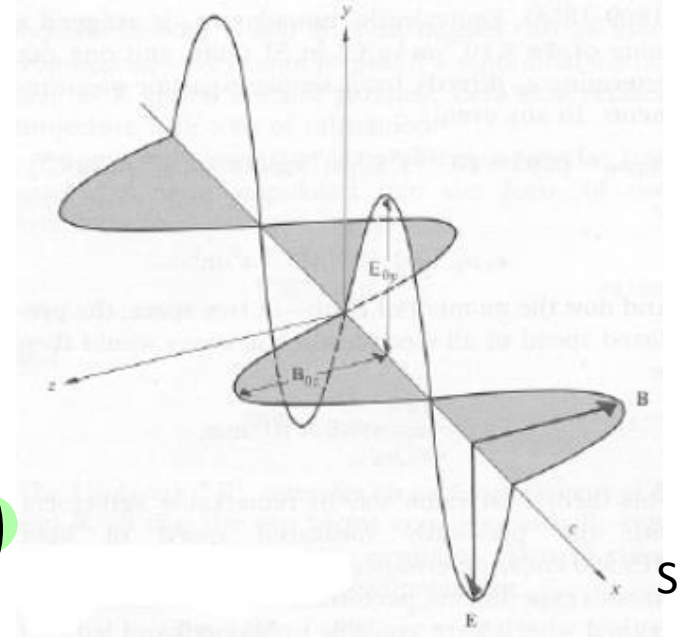
- Consider a harmonic, linearly polarized plane wave traveling through free space in the direction of $\vec{k}$

$$\vec{E} = \vec{E}_0 \cos(\vec{k} \cdot \vec{r} - \omega t)$$

$$\vec{B} = \vec{B}_0 \cos(\vec{k} \cdot \vec{r} - \omega t)$$

$$\vec{S} = c^2 \epsilon_0 \vec{E}_0 \times \vec{B}_0 \cos^2(\vec{k} \cdot \vec{r} - \omega t)$$

This is the instantaneous flow of energy
per unit area per unit time.



Irradiance and power

The average energy per unit area per unit time (irradiance I):

$$\begin{aligned} I = \langle S \rangle_T &= c^2 \epsilon_0 \left| \vec{E}_0 \times \vec{B}_0 \right| \left\langle \cos^2 \left(\vec{k} \cdot \vec{r} - \omega t \right) \right\rangle \\ &= \frac{c^2 \epsilon_0}{2} \left| \vec{E}_0 \times \vec{B}_0 \right| = \frac{c \epsilon_0}{2} E_0^2 \end{aligned}$$

The irradiance is proportional to the square of the amplitude of the electric field.

E is considerably more effective at exerting forces and doing work on charges than is B, we refer to E as the optical field and use the above equation almost exclusively.

The optical power (P): the time rate of flow of radiant energy (unit watts):

$$I = \frac{P}{S}$$

The inverse square law

From the solution of the differential wave equation
Spherical wave function:

$$E(r, t) = \left(\frac{\mathcal{A}}{r} \right) \cos(kr \mp \omega t)$$

From the energy conservation

$$I_1 \pi r_1^2 = I_2 \pi r_2^2$$

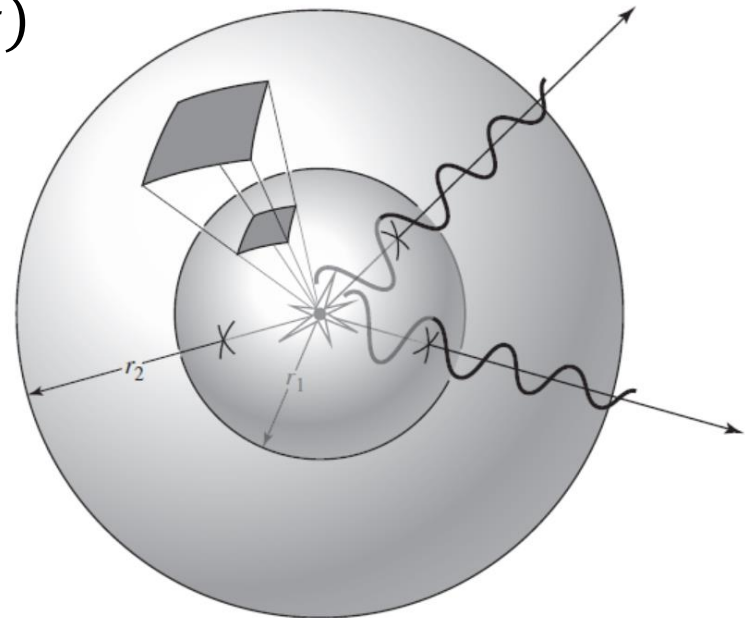
We get

$$r_1 E_0(r_1) = r_2 E_0(r_2)$$

As to arbitrary r

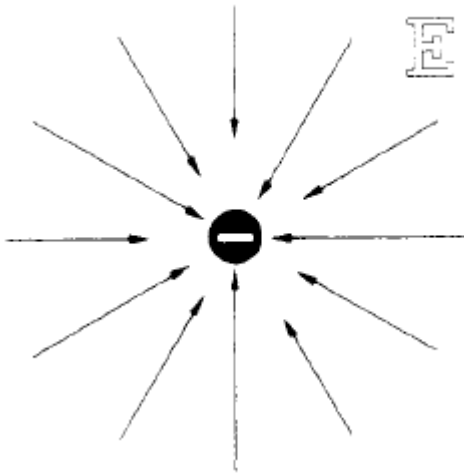
$$r E_0(r) = \text{constant}$$

$$I \propto \frac{1}{r^2} \quad \text{Inverse square law}$$



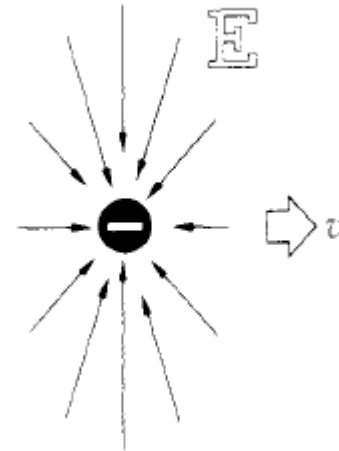
2.4 Radiation

A stationary charge: constant $\vec{E}$ -field, no $\vec{B}$ -field, **no radiation**.



Electric field of a stationary electron

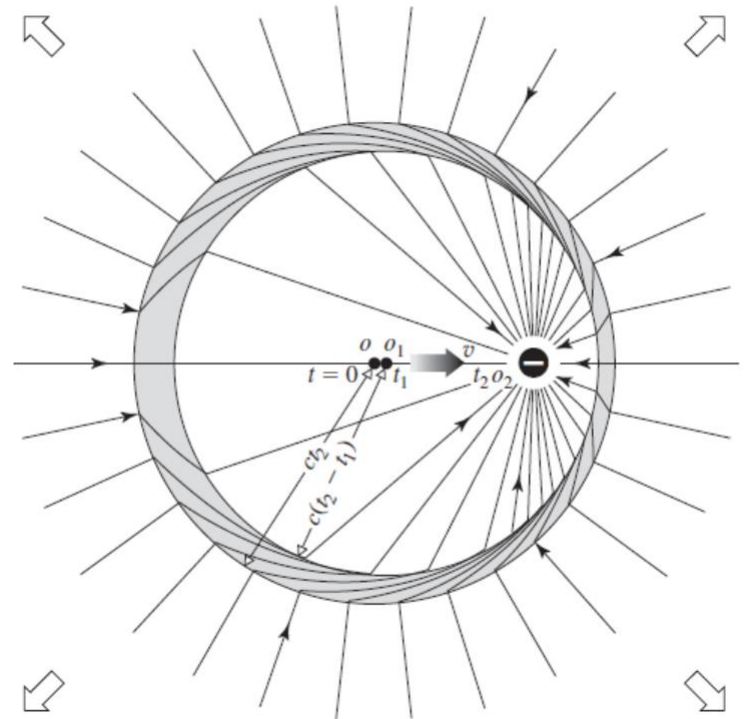
A uniformly moving charge: constant $\vec{E}$ -field and $\vec{B}$ -field, but **no radiation**.



Electric field of a moving electron
(v is comparable with c)

A nonuniformly moving charge: **radiation**.

- ❑ From this kinked region we see that there exists a transverse component of the electric field, $\mathbf{E}_T$.
- ❑ This transverse component propagates outward, away from the charge, at the speed of light.
- ❑ Thus, the pulse which contains the kink is not only a function of space, but also of time.
- ❑ From Maxwell's equations, we see that the transverse field is therefore accompanied by a magnetic field.



2.5 Photon

❑ The electromagnetic radiation is quantized and exists in the elementary amounts that we call **photons**.

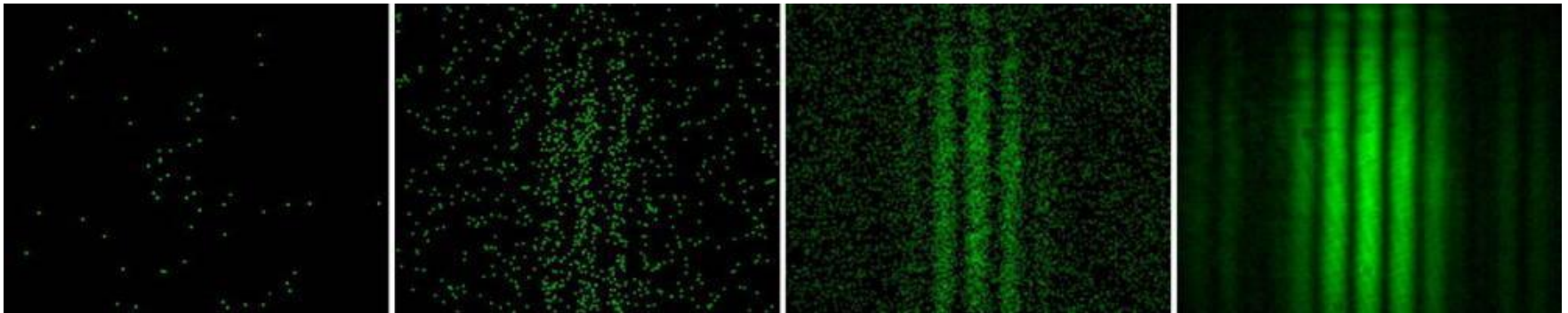
❑ The energy of single photon can be expressed as $E=h\nu$

Irradiance is proportional to the number of photon.

Quantum picture of light



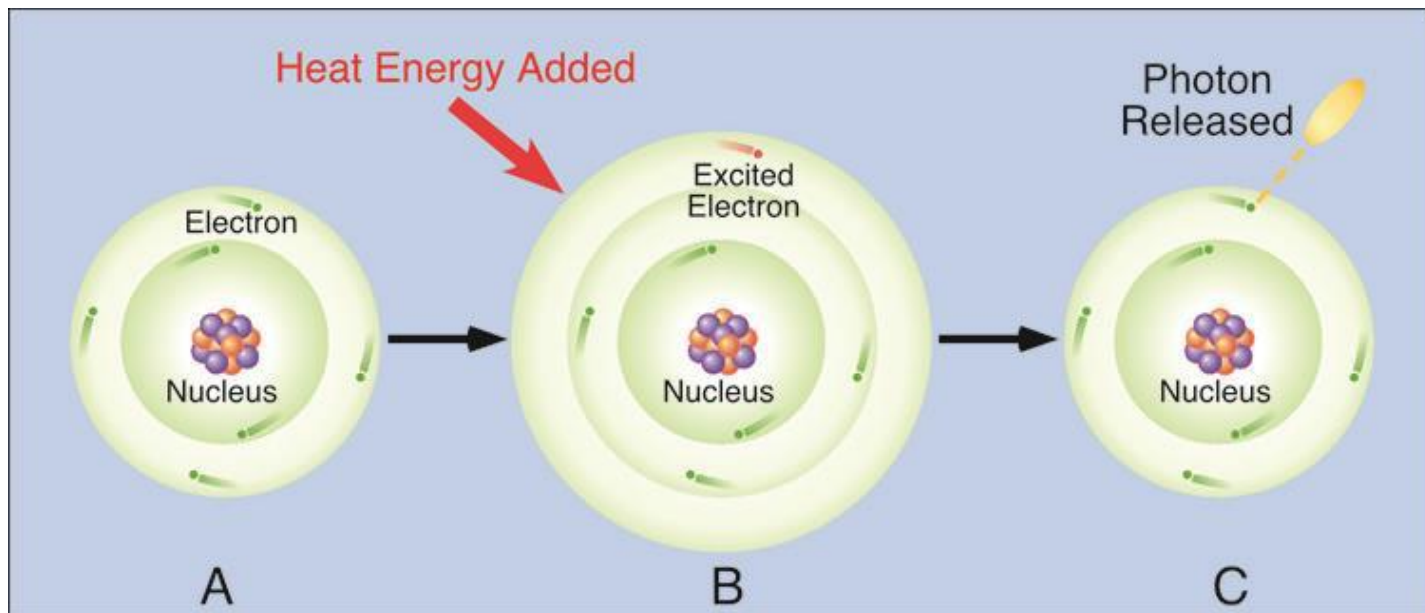
its wave property



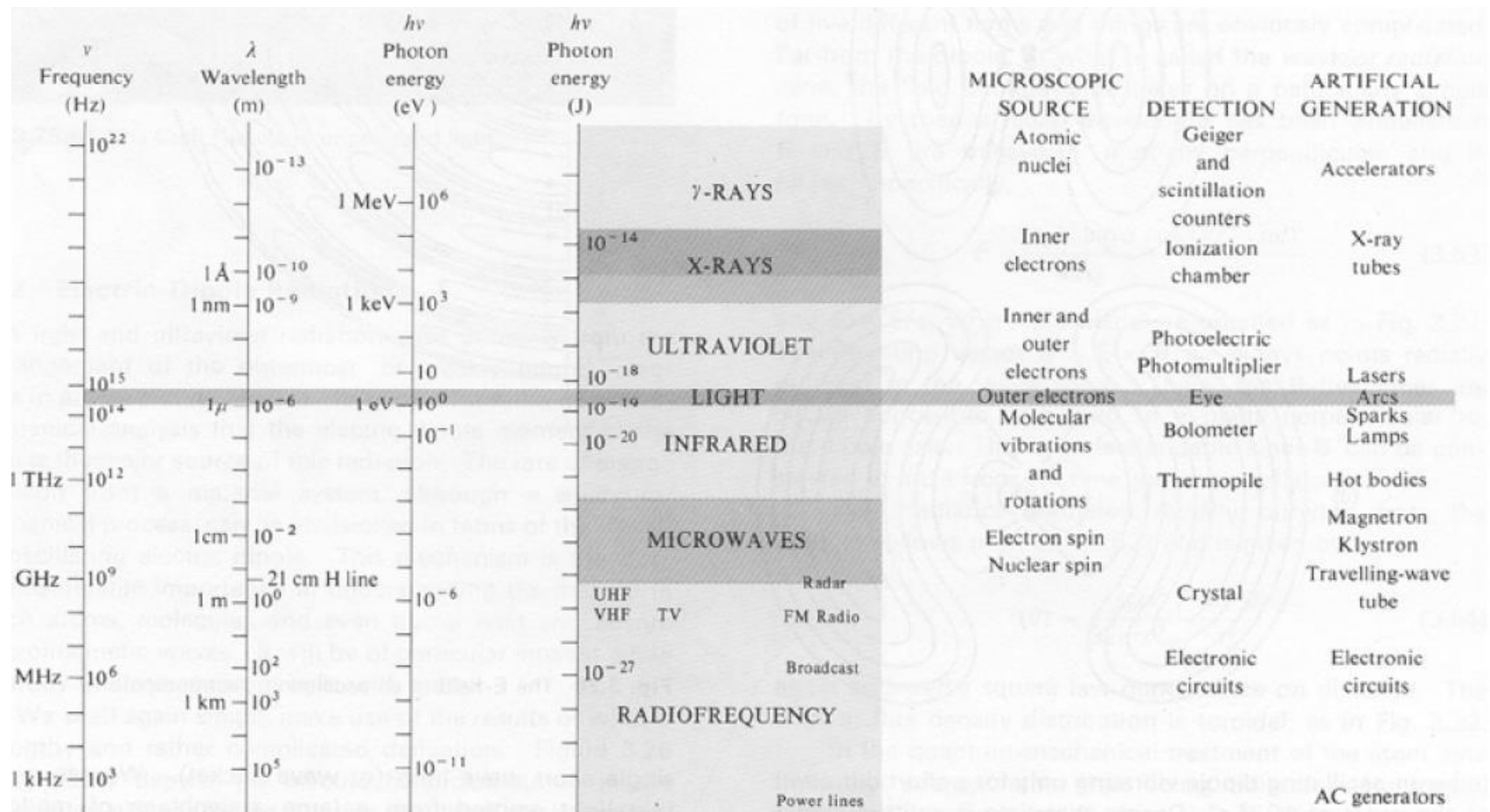
Single-photon camera recording of photons from a double slit illuminated by very weak laser light. Left to right: single frame, superposition of 200, 1'000, and 500'000 frames.

Emission of light from atoms

- ❑ (A) Without any disturbance, every electron is in the lowest possible energy state (**ground state**) available to it.
- ❑ (B) After absorbing a blast of energy, one or more electrons are excited to a higher energy level (**excited state**).
- ❑ (C) The excited state is unstable and temporary, electron drops back and returns to the ground state, with the emission of a **photon**.



2.6 The electromagnetic spectrum



2.7 Light in bulk matter

□ Objective: dielectric or nonconducting materials

The phase speed in the medium

$$v = 1/\sqrt{\epsilon\mu}$$

Absolute index of refraction (n):

$$n \equiv \frac{c}{v} = \frac{\sqrt{\epsilon\mu}}{\sqrt{\epsilon_0\mu_0}}$$

Consider the nonmagnetic materials

$$n = \sqrt{\frac{\epsilon}{\epsilon_0}}$$

The dependence of n on the wavelength of light is a well-known effect called **dispersion**.



Newton's prism

Scattering and absorption

- ❑ **Case I: the photon's energy matches that of one of the excited states.**
- ❑ As to the atom, the light will be absorbed, making a quantum jump to that higher energy level.
- ❑ In the dense atomic landscape of ordinary gases, solids and liquids, it is very likely that this excitation energy will rapidly be transferred, via collisions, to random atomic motion, thermal energy, before a photon can be emitted. It is referred to as **dissipative absorption (损耗吸收)**.

- ❑ **Case II: the photon's frequency or energy is lower than the resonance frequency.**
- ❑ The electromagnetic field of the light can drive the electron cloud into oscillation.
- ❑ Once the electron cloud starts to vibrate with respect to the positive nucleus, the system constitutes **an oscillating dipole** and will immediately begin to radiate at that same frequency.
- ❑ The resulting scattered light consists of a photon that sails off in some direction carrying the same amount of energy as did the incident photon—**the scattering is elastic.**

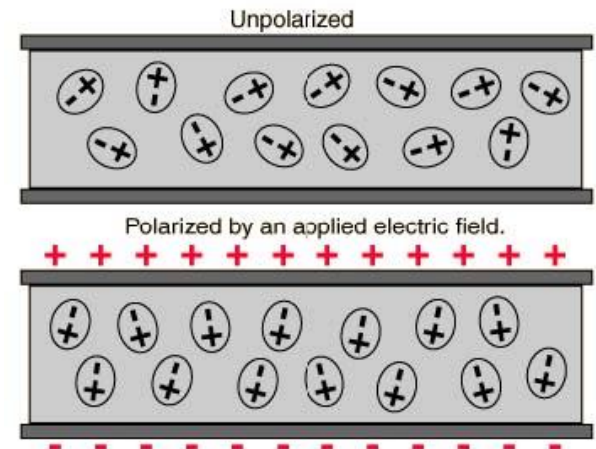
Dispersion 色散

- Dispersion corresponding to the phenomenon whereby the index of refraction of a medium is frequency dependent.

$$n = n(\omega)$$

- Electric polarization, P

$$\underline{(\epsilon - \epsilon_0)\vec{E} = \vec{P}}$$

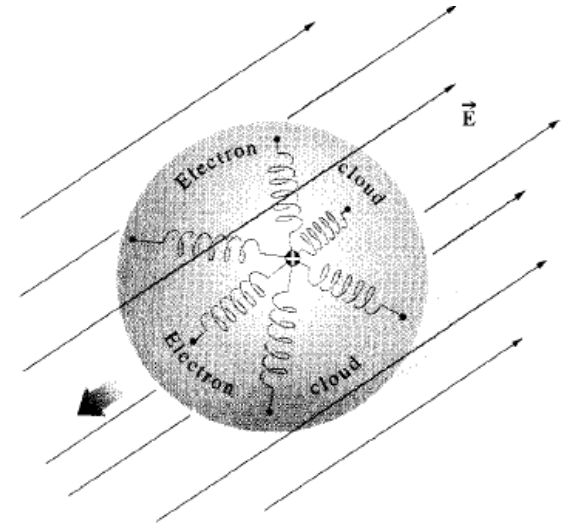


□ If the dielectric is subjected to an incident harmonic electromagnetic wave, its internal charge structure will experience time-varying forces.

□ Classical forced oscillator:

$$q_e E_0 \cos \omega t - m_e \omega_0^2 x = m_e \frac{d^2 x}{dt^2}$$

$$x(t) = \frac{q_e/m_e}{(\omega_0^2 - \omega^2)} E_0 \cos \omega t = \frac{q_e/m_e}{(\omega_0^2 - \omega^2)} E(t)$$



$$x(t) = \frac{q_e/m_e}{(\omega_0^2 - \omega^2)} E_0 \cos \omega t = \frac{q_e/m_e}{(\omega_0^2 - \omega^2)} E(t)$$

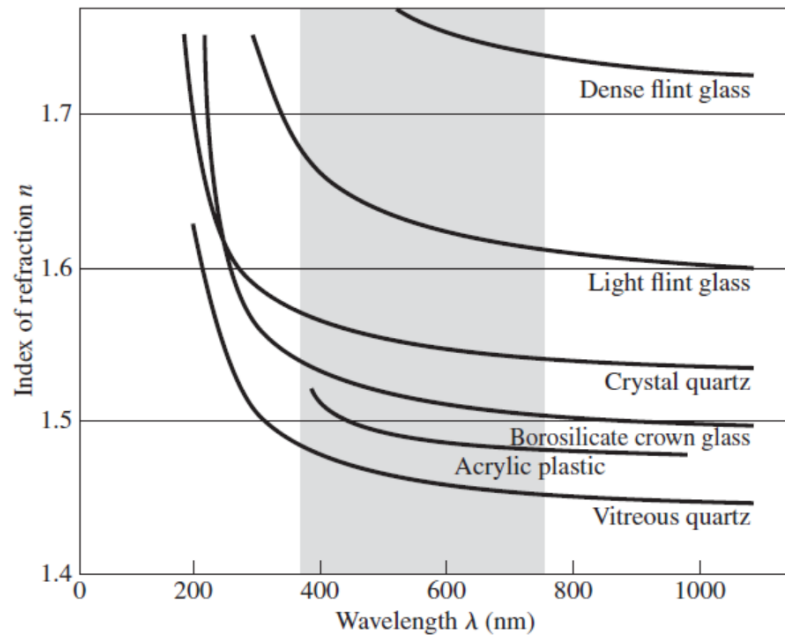
- ❑ Without the driving force, the oscillator will vibrate at its resonant frequency ω_0 ;
- ❑ If $\omega < \omega_0$; $x(t)$ and $E(t)$ have the same sign, meaning that the oscillator can follow the applied force, i.e., is in phase with it.
- ❑ If $\omega > \omega_0$; $x(t)$ and $E(t)$ have the opposite sign, meaning that the displacement $x(t)$ is in a direction opposite to that of the instantaneous force oscillator can follow the applied force $q_e E(t)$, i.e., is 180° out-of-phase with it.

❑ Dispersion equation:

$$P(t) = q_e x(t) N = \frac{q_e^2 N E(t) / m_e}{(\omega_0^2 - \omega^2)}$$

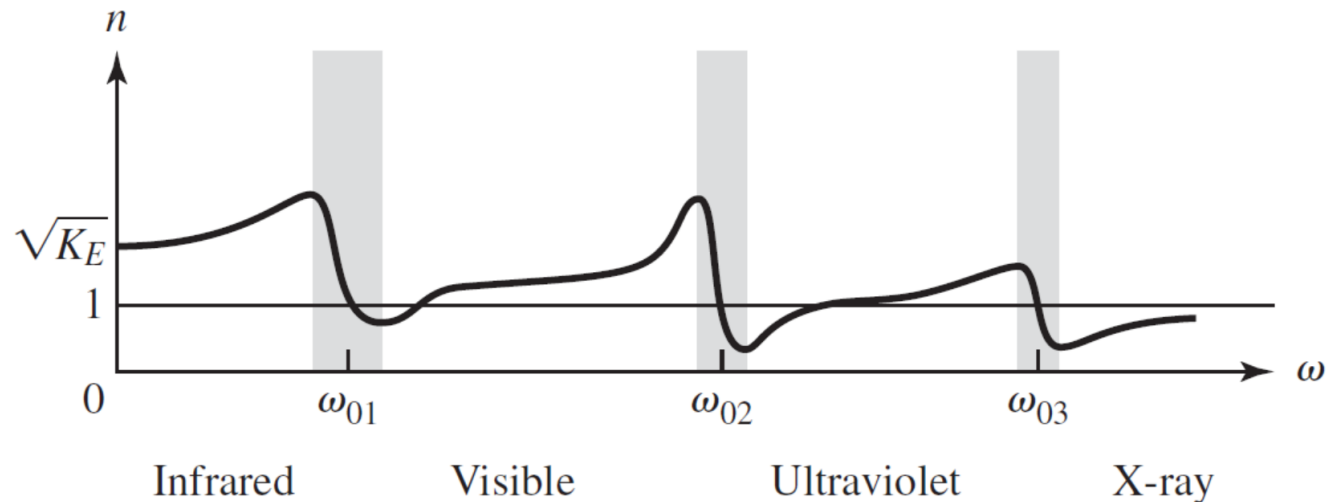
$$\epsilon = \epsilon_0 + \frac{P(t)}{E(t)} = \epsilon_0 + \frac{q_e^2 N / m_e}{(\omega_0^2 - \omega^2)}$$

$$n^2(\omega) = 1 + \frac{N q_e^2}{\epsilon_0 m_e} \left(\frac{1}{\omega_0^2 - \omega^2} \right)$$



$$n^2(\omega) = 1 + \frac{Nq_e^2}{\epsilon_0 m_e} \sum_j \left(\frac{f_j}{\omega_{0j}^2 - \omega^2} \right)$$

- ☐ Normal dispersion
- ☐ Anomalous dispersion



Homework

□ 3.1, 3.2, 3.6 , 3.20

3.1 Consider the plane electromagnetic wave (in SI units) given by the expressions $E_x = 0$, $E_y = 2 \cos[2\pi \times 10^{14}(t - x/c) + \pi/2]$, and $E_z = 0$.

- (a) What are the frequency, wavelength, direction of motion, amplitude, initial phase angle, and polarization of the wave?
- (b) Write an expression for the magnetic flux density.

3.2 Write an expression for the $\vec{E}$ - and $\vec{B}$ -fields that constitute a plane harmonic wave traveling in the $+z$ -direction. The wave is linearly polarized with its plane of vibration at 45° to the yz -plane.

3.6 The electric field of an electromagnetic wave traveling in the positive x -direction is given by

$$\vec{E} = E_0 \hat{j} \sin \frac{\pi z}{z_0} \cos(kx - \omega t)$$

- (a) Describe the field verbally. (b) Determine an expression for k . (c) Find the phase speed of the wave.

3.20 A 1.0-mW laser has a beam diameter of 2 mm. Assuming the divergence of the beam to be negligible, compute its energy density in the vicinity of the laser.